



December 2023

Duration: 3 Hours

Note:

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

20 Marks

1. **Rocks having silica as main constituent are called**
A) calcareous rock
B) siliceous rock
C) argillaceous rock
D) sedimentary rock
2. **The building stone can be dressed very easily**
A) any time
B) after seasoning
C) Just after quarrying
D) after some months of quarrying
3. **Ingredient of the brick earth which enables the brick to retain shape is**
A) magnesia
B) iron oxide
C) alumina
D) silica
4. **Maximum water absorption percentage for class "AA" Tile is**
A) 19
B) 24
C) 21
D) 22
5. **The age of the tree can be determined by**
A) annular rings
B) cambium layer
C) Medullary rays
D) heart wood
6. **The defect in wood due to hardening is**
A) twisted fibres
B) rind gall
C) burls
D) shakes
7. **Which cement contains high percentage of C_3S and less percentage of C_2S**
A) Rapid hardening cement
B) Quick hardening cement
C) Low heat cement
D) Expansive cement
8. **Alumina in excess in cement causes**
A) Shrinkage of cement
B) Increase in strength of cement
C) Expansion of cement
D) Reduction in strength of the cement
9. **For a rectangular foundation of width b , the eccentricity of the load should not be greater than**
A) $b/6$
B) $b/3$
C) $b/5$
D) $b/4$
10. **The type of foundation in which a cantilever beam used for joining the two footing is**
A) strap footing
B) raft footing
C) combined footing
D) pile footing
11. **IRC recommended Seating Load in Plate load test is**
A) 0.7 kg/m^2
B) 0.7 t/m^2
C) 0.7 g/m^2
D) 0.07 g/m^2
12. **_____ in which the base area is covered by heavy reinforced cement concrete is known as**
A) combined footing
B) strip footing
C) strap footing
D) raft footing
13. **Type of masonry in which the stones of irregular size and shapes are used and there are no regular courses is called**
A) Uncoursed Rubble Masonry
B) Rubble Masonry
C) Coursed Rubble Masonry
D) Ashlar Masonry

14. When a brick is Cut off length wise are called
 A) Bat B) Corbel
 C) King Closer D) Queen Closer
15. The Cantilever Projection provided at the entrance of the building is called
 A) Chejja B) Balcony
 C) Canopy D) Lintel
16. A is a horizontal member which is placed across an opening to support the position of the structure above it
 A) Plinth B) Lintel
 C) Sill D) Chejja
17. A bay window is a multi-panel window, which
 A) Is provided at corners B) Runs parallel to the wall
 C) Embeds inside the wall D) Projects outside the wall
18. The middle horizontal member of the shutter is called
 A) Lock Rail B) Median rail
 C) Mid rail D) Sill rail
19. The vertical portion between each tread on the stair is called
 A) Going B) Nosing
 C) Riser D) Winder
20. Baluster in a stair is themember supporting
 A) Vertical, landing B) Vertical, handrail
 C) Horizontal, handrail D) Horizontal, landing

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) i) Explain the Physical and Chemical classifications of rocks. ii) Write a note on factors affecting the deterioration of stones.	8	L2	1	1
	b) Explain the following industrial timber briefly with their uses. i) Veneers ii) Wood board	8	L2	2	1
2.	a) Explain in detail any three defects due to Fungi and two defects due to conversion in Timber.	8	L2	2	1
	b) List and explain characteristics of a good tile.	8	L2	1	1
3.	a) What are Bogue's compounds? Explain their role in setting and hardening process of cement.	8	L2	2	1
	b) With the help of a neat sketch, explain the working of Bull's trench kiln for manufacture of bricks.	8	L2	1	1
Unit – II					
4.	a) Compare English bond and Flemish bond (any 5). Explain polygonal rubble masonry with a neat sketch.	8	L2	4	1
	b) Explain the following with neat sketches: i) Strap footing, ii) Mat footing	8	L2	3	1
5.	a) What is safe Bearing Capacity of Soil? With the help of a neat sketch explain plate load test to determine the SBC of soil (Gravity loading method).	8	L2	3	1
	b) Explain timber lintel and brick lintel with a neat sketch.	8	L2	4	1
6.	a) Define the following with the help a neat sketch with respect to an arch. i) Soffit, ii) Extrados, iii) Voussoirs, iv) Spandril, v) Key.	8	L2	4	1
	b) List and explain the methods adopted for increasing the bearing capacity of soil.	8	L2	3	1

Unit – III

- | | | | | | | |
|----|----|---|----|----|---|---|
| 7. | a) | i) Explain the essential components of a floor with neat sketch. | 10 | L2 | 5 | 1 |
| | | ii) List any five characteristics of an ideal paint. | | | | |
| | b) | Define the following components of the staircase. | | | | |
| | | i) Newel post ii) Scotia iii) Winder iv) Tread v) Landing vi) Nosing | 6 | L1 | 5 | 1 |
| 8. | a) | Explain the procedure of applying the paints on: | | | | |
| | | i) Painting on New wood wok ii) Painting old plastered Surface. | 10 | L2 | 5 | 1 |
| | b) | Explain with the help of a sketch the purpose of providing Bay windows. | 6 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
